

Model documentation

The authoritative record of how the family office's asset-allocation study model is designed, what it assumes, where its limits lie, and how it has evolved.

PURPOSE	BALANCE SHEET	AS OF	SPEC OF RECORD
Research & study tool	Gen 3–5 single-family office	July 14, 2026	SPEC.md (authoritative)

SECTION I

Overview

The model is a modular asset-allocation model that lets the family-office team study three coupled questions on a single, integrated quarterly cash-flow ledger:

- 1. **Public-asset allocation** — strategic policy, constraints, and rebalancing.
- 2. **Spending & liquidity** — withdrawal rules, reserve floors, and coverage.
- 3. **Private-equity pacing** — capital calls, distributions, and how value evolves.

Its outputs are reports and data files — a ledger, a written report, a run manifest, and a scenario comparison. It is deliberately not a production trading, accounting, or order-management system; it is a tool for thinking clearly about the questions above.

IN PLAIN TERMS

Everything the model does is recorded as dated entries on one shared ledger — the same discipline as a household's books — so any result can be traced back to the assumptions and cash flows that produced it.

SECTION II

The standing principle

This model is built for a Gen 3–5 single-family office balance sheet — one that typically holds large illiquid private real estate, operating-company interests, development assets, and land alongside its public portfolios. That context is load-bearing for every modeling decision: assumptions that suit an endowment or a mass-market retirement plan often fail here.

Four disciplines shape every spending, liquidity, and real-estate decision that follows — each less a slogan than a working rule about the gap between what an asset is worth and what it can fund:

A larger balance sheet is not a larger checkbook.

Net asset value can climb for years without producing a dollar the household can actually spend.

An appraisal sets a mark, not a budget.

The figure a private asset is carried at is a valuation, not cash waiting to be drawn against.

Land and development pay in the future tense.

Appreciation on land or a project in progress is real, but it is not income arriving this year.

Owning the company is not owning its cash.

Operating-company value only reaches the portfolio once a distribution is actually declared.

When a phase of work appears to treat total net asset value as a stand-in for spendable resources — or appraisal value as cash, or operating-company value as liquid — that assumption must be named explicitly, and either justified for its narrow purpose or queued as a separate piece of work.

Distinctions the model must keep separate

CONCEPT	WHAT IT MEANS FOR THE HOUSEHOLD
Total NAV	Every modeled asset summed, including illiquid appraisal carry. Currently the only NAV concept the model has.
Liquid NAV	Public sleeves plus cash — immediately tradable. Made load-bearing for rebalancing in Phase 8.
Income-producing NAV	Assets generating distributable yield. Real estate carried at appraisal but paying nothing is not income-producing for spending.
Distributable income	Actual cash reaching the household this period — distinct from NAV growth or appraisal step-ups.
Locked / illiquid NAV	Private real estate, operating-company equity, development, land. May appreciate, but cannot fund spending without a liquidity event.
Spendable resources	What can actually be spent this period without forced sales. Not yet a model concept — the subject of ongoing work (L19).

SECTION III

How the system is built

There is one load-bearing object — the quarterly ledger — and everything else is either a producer or a consumer of rows on it. Configuration and scenarios resolve into a single study configuration; an orchestrator walks quarter by quarter, asking each component to contribute its flows; the ledger sorts, chains, and validates every row before anything is written out.

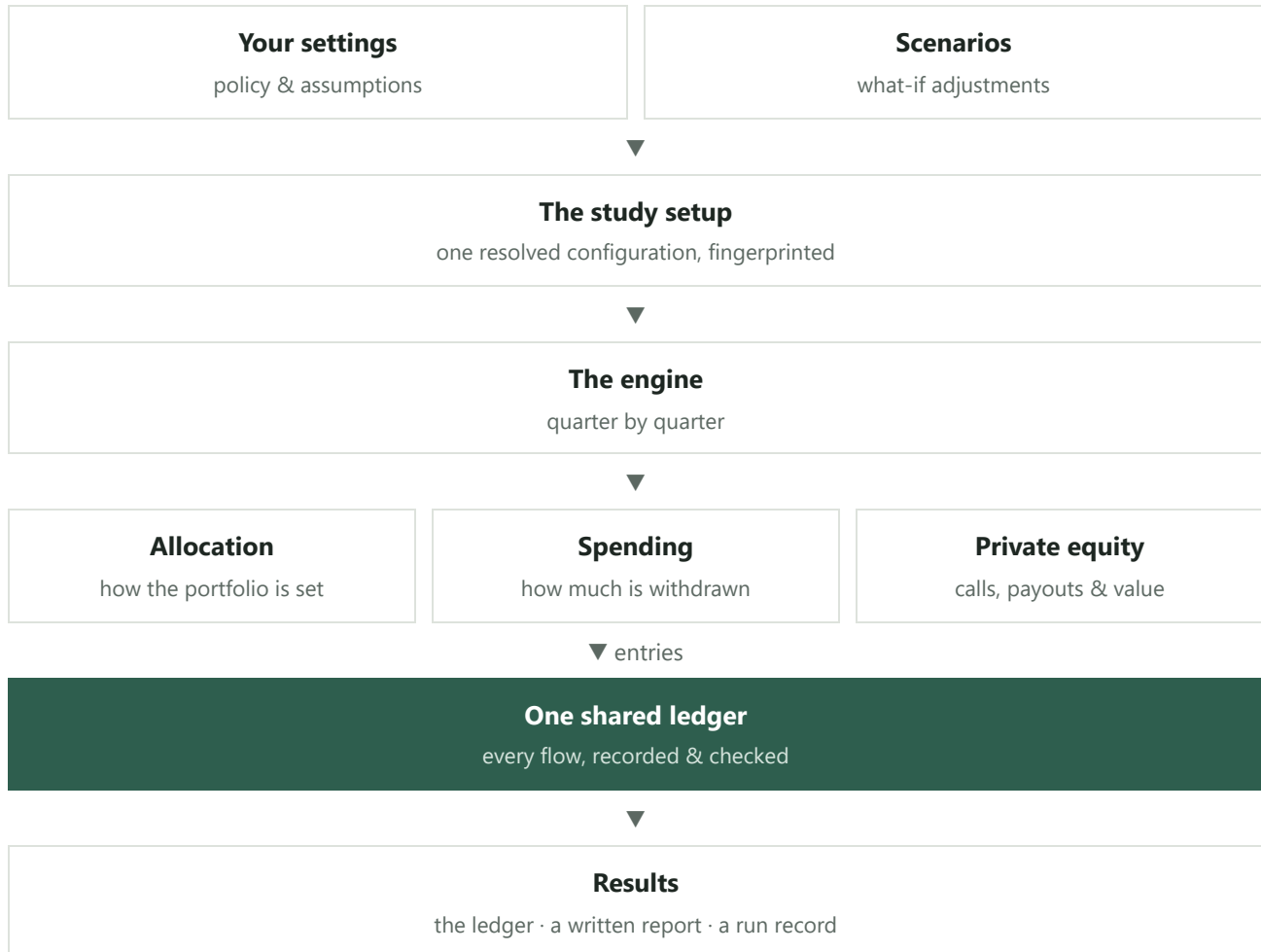


Figure 1 — One shared ledger sits at the center; each part of the model adds to it, and every result is drawn from it.

Reproducibility

Every run writes a manifest recording the configuration fingerprint, the data fingerprint, library versions, the random seed, timestamps, and the list of files produced. Because fingerprints are computed from the resolved configuration itself, an adjustment made in memory hashes identically to the same edit made on disk. The run identifier itself is validated to be path-safe, so a caller-supplied identifier can never place outputs outside the run directory.

THE GUARANTEE

Run the same configuration twice and you get two separate, timestamped output folders whose ledger contents are byte-identical once the run identifier is set aside. Results are never overwritten, and nothing is left to chance.

SECTION IV

How the books stay balanced

A run is not allowed to finish unless every one of the checks below holds. They are enforced automatically at the end of every run — the accounting equivalent of a balanced set of books.

GUARANTEE	WHAT IT ENSURES
Consistent ordering	Within each quarter and account, flows are recorded in one fixed order — income, return, calls, distributions, marks, spending, rebalance, then cost.
Each row balances	Ending value equals starting value plus the flow — and each row's start matches the prior row's end, so the chain never breaks.
Quarters tie out	Every account's flows for a quarter sum exactly to its change in value, and external cash (income, spending, cost) matches the household's net wire.
Internal moves net to zero	Rebalancing and private-equity calls / distributions move money between accounts without creating or destroying it.
Paired PE legs	Every capital call and distribution has a matching, opposite entry against cash — catching a missing or sign-flipped leg.
Value moves honestly	Total portfolio value changes only through market performance and external cash — never through an internal transfer.
No missing numbers	No blank or invalid values anywhere, and a single run identifier per ledger.

A DELIBERATE CHOICE WORTH NOTING

Transaction costs are treated as money leaving the household to a third party, not as internal leakage. This keeps the "value moves honestly" guarantee intact and means reported returns are already net of costs.

What's inside

Allocation

Two interchangeable allocators sit behind a common interface. A reference allocator simply returns the target weights written in the policy configuration — useful, transparent, always valid. An optimizing allocator (opt-in) solves a mean-variance problem against explicit capital-market assumptions with long-only bounds. The two are not meant to produce the same numbers; they are held to the same structural standards instead.

Rebalancing

A zero-cost rebalancer trades exactly the gap between current and target allocations. A cost-aware rebalancer (opt-in) applies a straightforward per-trade cost on traded volume. Market impact, slippage, and execution delay are intentionally not modeled at this stage.

Spending & liquidity

Three withdrawal rules are available, each precomputed at the start of a run and inflation-aware: **flat-real** — a steady annual amount, split evenly across quarters, stepped up for inflation each year, with an optional floor and ceiling; **smoothing** — the same target, but eased toward gradually so spending changes are gentle rather than abrupt; and **Owl (guardrails)** — a Guyton-Klinger guardrail rule that ratchets spending up or down when the withdrawal rate drifts outside a set band.

Alongside spending, the model derives liquidity metrics after each run — how many months of spending the liquid assets cover, how often coverage falls below a floor (18 months by default), and the worst peak-to-trough decline.

Private-equity model

A Takahashi-Alexander cash-flow projector schedules each fund's capital calls, distributions, and value marks from vintage over its life. Calls follow a contribution schedule during the commitment period; distributions follow a "bow" curve that accelerates as the fund matures — and in the final projected quarter the fund is fully liquidated, so every fund terminates at exactly zero value. A pinned golden result guards the arithmetic against drift.

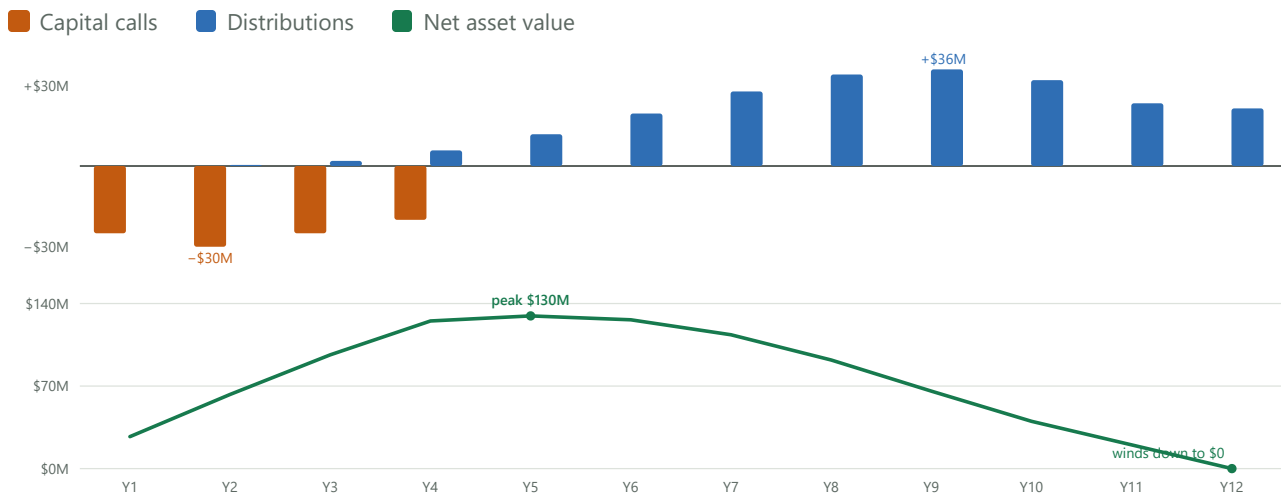


Figure 2 — A single \$100M fund on the model's pinned defaults (the golden result). Top: annual cash flows — calls drawn below the line, distributions above. Bottom: net asset value follows the classic hill from vintage to a forced full wind-down at zero in year 12.

Scenarios

Scenarios are lightweight overrides applied on top of the base configuration; the engine never branches on a scenario's name. Five canonical perturbations ship with the model:

SCENARIO	WHAT IT STRESSES
Base	Reference run, no override.
Public drawdown	A 25% public-equity fall with a four-quarter recovery.
Delayed PE distributions	Distributions arrive later in each fund's life.
Clustered calls	Capital calls concentrated early in the commitment period.
Inflation shock	Annual inflation lifted from 2.5% to 6.0%.

SECTION VI

What the model assumes

Every model rests on simplifications. These are stated plainly so results are never over-read:

Quarterly grain

All flows and returns are quarterly; no within-quarter dynamics.

Deterministic returns

Return paths are set, not random; a scenario hardcodes its path.

Scheduled PE growth

PE value grows at a constant rate; distributions follow a curve, not market events.

Independent buckets

Asset classes are treated as statistically independent at this stage.

No taxes

No tax-lot accounting, after-tax returns, or charitable structures yet.

Single currency

US dollars only.

A further early-stage assumption — that private equity can be rebalanced as freely as public holdings — was corrected in Phase 8 and is now handled by an illiquidity overlay (see the limitations that follow).

Known limitations

Each limitation is stated openly so a reader cannot misinterpret a result. The register below is the single source of truth for status; it separates what the code does from how the output should be read.

#	TOPIC	STATUS
L1	PE timing scenarios mechanically affect returns	Partially resolved
L2	Returns are level-dependent, not regime-dependent	Engine shipped
L3	Reference vs optimizer weights not directly comparable	Accepted
L4	Optimizer's default assumptions were a placeholder	Resolved
L5	PE-leg pairing key is fragile	Open · schema
L6	Correlation-shock scenario was omitted	Resolved
L7	Smoothing with zero weight freezes spending	Accepted
L8	Rebalancer treated PE as a liquid sleeve	Resolved
L9–L12	Third-party install & solver constraints (optimizer)	Accepted / note
L13	Cost-aware allocation had no path dependence	Resolved
L14	Only linear transaction cost is modeled	Partially resolved
L15	Guardrail reacted to forecast, not realized NAV	Resolved
L16	Guardrail was scale-invariant in initial NAV	Resolved
L17	Cross-engine metric comparisons aren't meaningful	Accepted
L18	Guardrail misread inflation shock as headroom	Resolved
L19	Spending-base realism for illiquid balance sheets	Partially resolved

Where the register stands

19

entries — Resolved 8 · Partially resolved 3 · Accepted / environment note 7 · Open 1

1

genuinely open item (L5), sequenced on the roadmap

L19

the most client-relevant partial — see below

What the statuses mean. *Resolved* — a code change closed the concern; the entry is kept for the audit trail. *Partially resolved* — closed under one engine or scope, still open under another. *Accepted* — documented and expected; no change planned (often a third-party constraint or an interpretation point). *Open* — genuine future work, sequenced on the roadmap.

THE ONE TO WATCH — L19

The guardrail rule still measures the withdrawal rate against total net asset value. For a family office holding large private real estate, operating-company equity, development, and land, total NAV can materially overstate what is actually spendable. Making spending rules aware of a liquidity-adjusted base is the most client-relevant item on the backlog — and the direct expression of the standing principle at the front of this document.

How it is validated

Confidence in the model rests on a broad automated test surface and a strict reproducibility check — not on trust.

550

automated tests passing at the current head (grown from 71 at the core's first freeze)

2×

every integration run repeated & compared byte-for-byte

≥ 80%

coverage gate on core modules (95.9% at last measurement); lint clean, enforced at push

Tests fall into three families: construction-level checks guaranteed by the ledger's own build step; validation-level checks that reject any run whose books don't balance; and cross-scenario checks confirming that one scenario's results never leak into another's. On every integration run, the study is executed twice and the two ledgers are compared — any drift signals that determinism has been lost.

How it has evolved

The model has grown in deliberate phases, each with a frozen design before any code lands. The arc below condenses that history.

PHASE	MILESTONE
1–3	Foundations — the ledger, invariants, allocators, rebalancers, spending rules, and PE model.
4	Path-dependence done safely — rules react to realized, not forecast, state; cost-aware allocation.
5–6	Explicit capital-market assumptions; a scenario-driven correlation shock.
7–8	The STAIRS PE adapter; the illiquidity overlay that stops rebalancing private equity.
9–11	Manager & fund metadata; transaction-cost diagnostics; scale-aware guardrails.
12–14	Spending-base machinery; a distributable-income flow; the RE/OpCo producer and cash-flow workbook ingestion.
15–17	Investment-summary / position ingestion; liquidity-coverage diagnostics; orchestration of the new layers.
18–22	Spending-base bridge; PE capital-call obligation bridge and worksheet reconciliation; policy gates; manager-terms diagnostics.
24	A first-class entity dimension — reproducing the firm's standard 11-tab allocation study for a single entity from source documents.

CHANGE DISCIPLINE

This document is doc-as-spec: every behavior change updates it in the same commit series, and a tracker-level governance check flags any that slips through. That check has been exercised in earnest once — three hardening fixes from an external review on May 5, 2026 shipped without their entries, were flagged the same day, and were documented in a governance follow-up on July 14, 2026. The register carries the audit trail.

Recent work

Monte Carlo & stochastic stress

A simulation layer for synthetic stress testing is built and tested — seeded, reproducible path generation feeding liquidity-coverage and breach detection, with required reserves computed by an exact solve rather than an approximation. It is advisory only and explicitly not decision-grade until calibrated against real client data.

Entity dimension (Phase 24)

The model now carries a first-class entity dimension and can reproduce the firm's standard 11-tab asset-allocation study for a single entity, deterministically, from documents it already ingests — validated to the penny against a real trust workbook, and re-validated in July 2026 against a mixed-vintage revision of the same workbook. Client-specific material stays out of the shared codebase; only generic, methodology-level code is committed.

Market-data ingestion

Morningstar Direct index returns now flow into the capital-market-assumption layer through a file-based ingestion path, giving the optimizer's assumptions an empirical calibration anchor.

Hardening (external review)

A May 2026 external review produced three fixes now part of the model's contracts: run identifiers are validated path-safe before they name output directories; the configuration fingerprint covers every run-affecting input, including the ingestion and gate layers; and the private-equity projector, call diagnostics, and worksheet reconciliation each closed an edge case (terminal wind-down to zero, fund counting, zero-by-zero division).

What comes next

The near-term sequence, in order:

1. **Cash-flow classification (L19).** Complete the classification of worksheet rows so the spending base reflects genuinely distributable income — the immediate gate.
2. **PE pacing improvement (Phase 23).** Bring in fund-level commitment data while staying deterministic and reconciled to the worksheet. Design is locked; implementation awaits real commitment-book data.
3. **Distribution & income realism.** Full recurring / non-recurring classification once row classification is stable.
4. **Manager-terms enrichment.** Populate lockups, notice periods, gates, and fee terms for the diagnostics layer.

Improvements in progress, on a longer horizon: semi-liquid redemption modeling, secondary-sale liquidity haircuts, fee- and tax-aware cash-flow modeling, entity-governance restrictions, and regime-dependent returns.

This document records implementation behavior and is updated as the model changes. It is authoritative for how the model is built and behaves; the specification and project-scope statements govern intent and purpose. Source of record:

docs/MODEL_DOCUMENTATION.md in the model repository; this export was generated from it on July 14, 2026. Figures are drawn from the model's own pinned golden results.